

5 Photosynthesis is the energy transfer process that occurs in chloroplasts.

Fig. 5.1 shows some biochemical events that occur in a chloroplast during the light-dependent stage of photosynthesis. Photosystems I and II (PSI and PSII) and some associated proteins of the thylakoid membrane are shown.

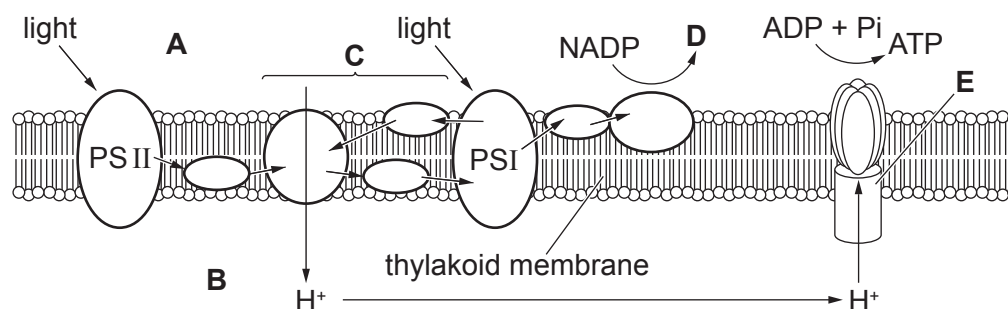


Fig. 5.1

(a) (i) **A** and **B** are areas of the chloroplast. Name areas **A** and **B**.

A

B [2]

(ii) The group of proteins labelled **C**, PSI and the protein labelled **E** are involved in a specific biochemical process during the light-dependent stage of photosynthesis.

Name this specific biochemical process and the protein labelled **E**.

process

E [2]

(iii) Product **D** is used during the Calvin cycle.

Identify product **D** and describe its specific role in the Calvin cycle.

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- (b) The rate at which plants use photosynthesis to transfer light energy into chemical energy that is stored in plant biomass can be measured in megajoules per square metre per year ($\text{MJ m}^{-2} \text{yr}^{-1}$).

Fig. 5.2 compares the rates of energy transfer for different ecosystems.

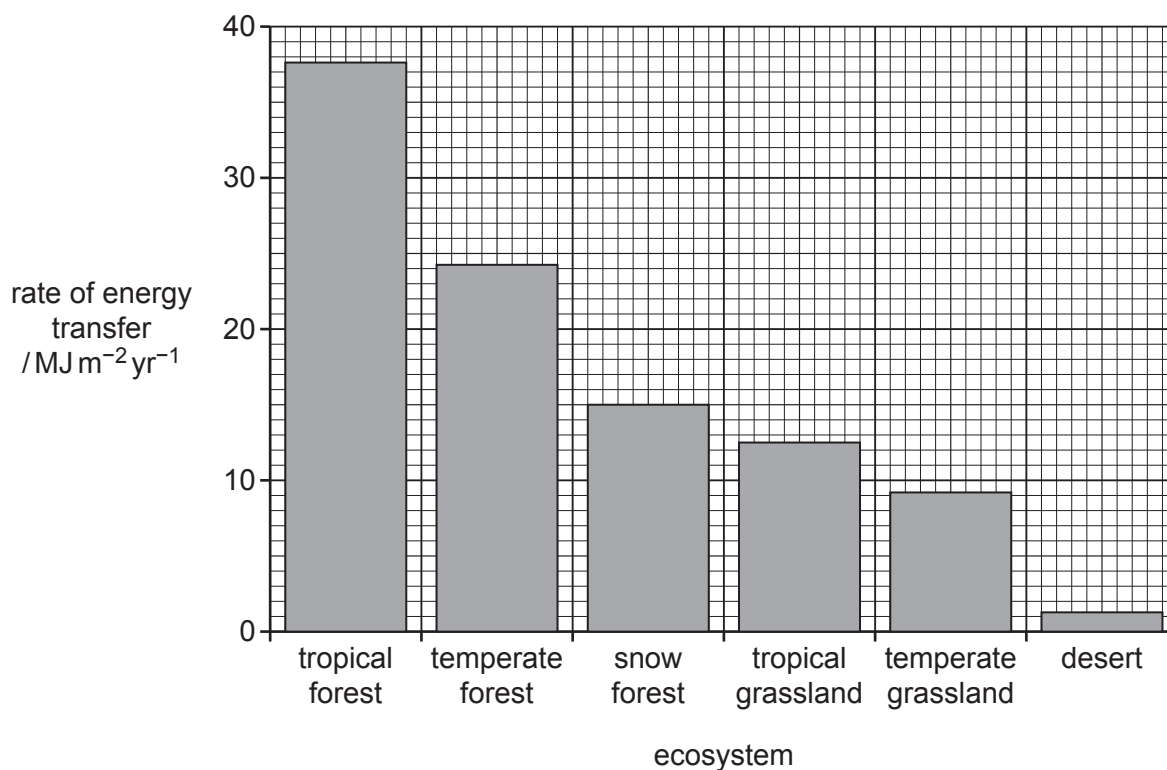


Fig. 5.2

- (i) Limiting factors affect the rate of photosynthesis.

Describe **and** suggest how the results for tropical forest, temperate forest and snow forest in Fig. 5.2 show the effect of limiting factors on photosynthesis.

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- (ii) Suggest reasons why grasslands and desert have a lower rate of energy transfer by photosynthesis than forests.

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[Total: 13]